

Data Preprocessing and Exploratory Data Analysis

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Outline

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Data Collection

- Data can originate from many sources: sensors, application logs, web scraping, surveys, or existing operational databases.
- The way it is collected determines its reliability; sampling bias, missing context, or inconsistent labeling introduced at collection time cannot be fully corrected later.
- Before any modeling can begin, this raw information must be organized into a machine-readable format.

Key quality dimensions to check at intake:

- Accuracy: values reflect the true underlying quantity
- Completeness: required fields are actually populated
- Consistency: the same fact is not represented differently across records
- Timeliness: the data is current enough for the task at hand

Data Representation Formats

- Structured: tabular/relational data with rows as examples and columns as features (spreadsheets, SQL tables)
- Semi-structured: JSON, XML, or log files with nested or optional fields, no fixed schema
- Unstructured: raw text, images, audio, video; typically converted into numerical feature vectors before use

Data as Vectors and Matrices

- A single example is a feature vector $\mathbf{x}_i \in \mathbb{R}^d$ with components $x_{i1}, \dots, x_{id}$, one per feature.
- A dataset of n examples is stored as a design matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$, with row i equal to $\mathbf{x}_i^\top$.
- Column j of $\mathbf{X}$ holds feature j across all examples; useful for computing per-feature statistics or scaling.
- Convention note: some texts transpose this, using $d \times n$ with examples as columns; check the source before mixing formulas.

Features and Labels

- Label vector $\mathbf{y}$: continuous $\mathbf{y} \in \mathbb{R}^n$ for regression, or discrete $y_i \in \{1, \dots, K\}$ for classification.
- Feature types: numerical (continuous or discrete), categorical nominal (unordered), categorical ordinal (ordered).
- Data is partitioned as $\mathbf{X}^{(\text{tr})}$, $\mathbf{X}^{(\text{val})}$, $\mathbf{X}^{(\text{te})}$, each with matching label subsets.
- Feature space: the d -dimensional space containing every $\mathbf{x}_i$; as d grows, data becomes sparse in this space (curse of dimensionality).

Data Preprocessing and Cleaning

- Collected data rarely arrives ready for modeling; it contains inconsistent formats, duplicate entries, or fields irrelevant to the prediction task.
- Cleaning resolves these structural issues before any statistical or geometric assumption of a learning algorithm is applied to the data.
- The specific steps required depend on the data types present and the model that will consume them.
- Deduplication: remove or merge repeated records from logging errors or merged sources
- Encoding: one-hot encoding for nominal categories, ordinal encoding for ordered categories
- Scaling: min-max normalization to $[0, 1]$, or standardization to $\mu = 0$, $\sigma = 1$
- Splitting into train/validation/test done before computing any

Handling Missing Values

- Missing entries arise from sensor failures, survey non-response, or incomplete records.
- Ignoring them silently can bias parameter estimates or shrink the effective training set.
- The right strategy depends on how much data is missing and why it is missing.
- Deletion: listwise or pairwise removal; reasonable when missingness is small and random
- Imputation: mean/median/mode, or model-based (k-NN, regression) for more structure
- Missingness indicator: add a binary flag so the model can use the pattern of missingness itself
- Mechanism matters: MCAR, MAR, and MNAR data call for different strategies

Handling Outliers

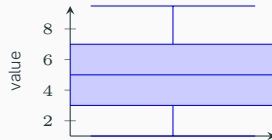
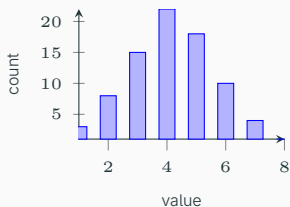
- Outliers are values far from the bulk of the data, arising from measurement error, data entry mistakes, or genuine rare events.
- Left unaddressed, they can dominate distance- or variance-based computations, skewing the fitted model.
- Not every outlier should be removed, so detection and treatment are handled as separate decisions.
- Detection: z-score thresholding ($|z| > 3$), IQR rule ($Q_1 - 1.5 \text{ IQR}$, $Q_3 + 1.5 \text{ IQR}$), or visual inspection
- Treatment: winsorizing/capping, log or Box-Cox transform to reduce skew, or removal if clearly erroneous
- Caution: some outliers are the signal, not noise (e.g., fraud detection, rare disease cases)

Exploratory Data Analysis: Goals

- Visual exploration is often the fastest way to understand a dataset before committing to a model.
- It surfaces patterns, relationships, and irregularities that summary statistics alone can miss.
- The goal is to let the data guide modeling choices, not to confirm assumptions made in advance.
- Check distribution shape, skew, and modality of individual features
- Detect relationships and redundancy between feature pairs
- Spot outliers and anomalies before they distort training
- Guide feature engineering and transformation decisions

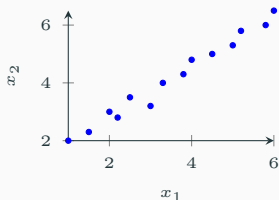
Visualizing Distributions

- Histogram: shows the frequency distribution of a single numerical feature
 - Use when checking skewness, modality, or the need for a transform
- Box plot: summarizes median, quartiles, and potential outliers via whiskers
 - Use when comparing spread and outliers across features or groups



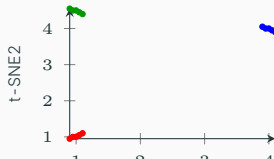
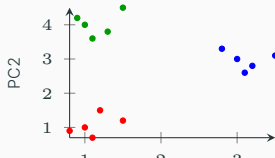
Visualizing Relationships

- Scatter plot: shows the relationship between two numerical features
 - Use when checking for linear/nonlinear trends or clustering between two features
- Correlation heatmap: shows pairwise linear association across many features at once
 - Use when screening many feature pairs for redundancy or multicollinearity



Visualizing High-Dimensional Data

- Dimensionality reduction projects high-dimensional features into 2D for visual inspection
- PCA: linear projection onto directions of maximum variance
 - Use when a fast, linear, variance-preserving view is enough
- t-SNE: nonlinear projection that preserves local neighborhood structure
 - Use when cluster structure matters more than global distances; slower, non-deterministic
- Both are used for visualization only, not as inputs to downstream models



Questions?
